Lecture 9: Large Language Models (LLM)

Chao ZHANG

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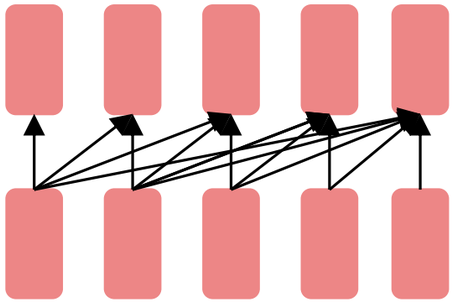
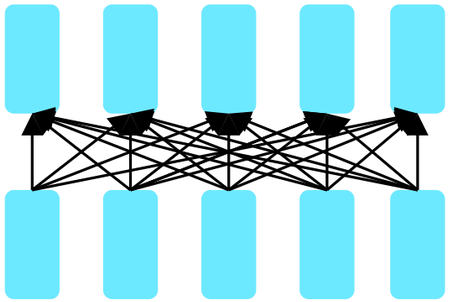
Three architectures for large language models



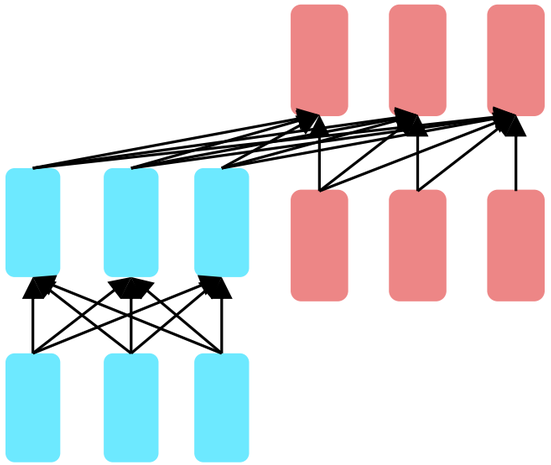
Decoders

GPT, Claude, Llama Mixtral



Encoders

BERT family, HuBERT



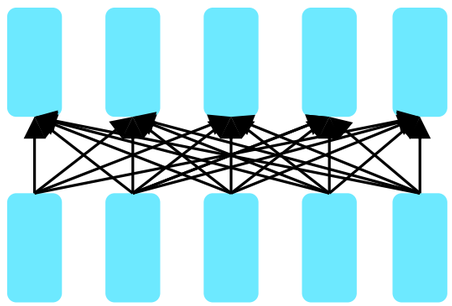
Encoder-decoders

Flan-T5, Whisper

LLMs

BERT vs GPT

Encoders

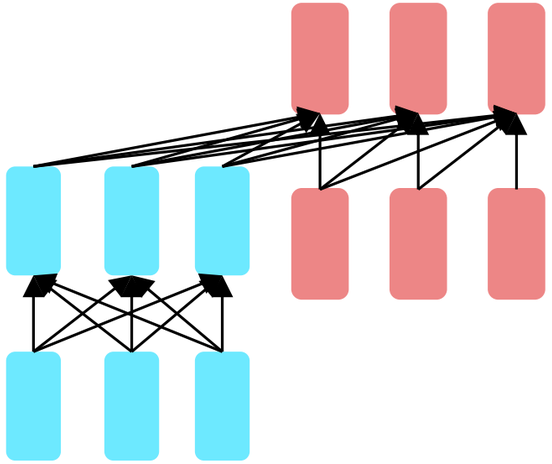


Many varieties!

- Popular: Masked Language Models (MLMs)
- BERT family

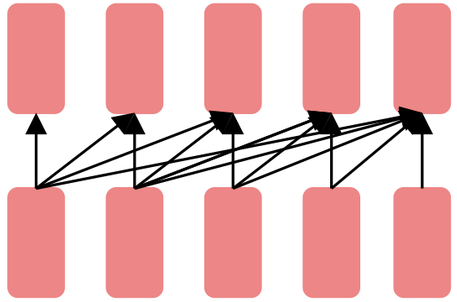
- Trained by predicting words from surrounding words on both sides
- Are usually **finetuned** (trained on supervised data) for classification tasks.

Encoder-Decoders



- Trained to map from one sequence to another
- Very popular for:
  - machine translation (map from one language to another)
  - speech recognition (map from acoustics to words)

This lecture: decoder-only models



Also called:

- Causal LLMs
- Autoregressive LLMs
- Left-to-right LLMs

- Predict words left to right

Empirical Observations (Wang et. al)

- Decoder-only models outperform encoder-decoder models using similar configuration

	EAI-EVAL	T0-EVAL
Causal decoder	**44.2**	**42.4**
Non-causal decoder	43.5	41.8
Encoder-decoder	39.9	41.7
Random baseline	32.9	41.7

Introduction

Learning Through Next-Word Prediction

Task Transformation Examples

What is Next-Word Prediction?

- Fundamental training task for large language models (LLMs).
- Models are trained to predict the next word given a sequence.
- This simple objective drives the model to capture complex language patterns.

Introduction

Learning Through Next-Word Prediction

Task Transformation Examples

Learning Language Structure and Patterns

- **Grammar & Syntax:** Models learn the rules of language.
- **Semantic Relationships:** Understanding how meaning is constructed.
- **Contextual Dependencies:** Capturing long-range relationships between words.

Capturing World Knowledge

- Trained on vast amounts of diverse texts.
- Learns factual information, cultural context, and common-sense reasoning.
- Embeds real-world knowledge into its representations.

Emergent Abilities and Efficiency

- **Self-Supervised Learning:** No need for explicit labels.
- **Generalization:** Models perform tasks they were not explicitly trained for.
- **Representation Learning:** Internal representations capture syntactic and semantic nuances.
- **Scalability:** Simple, quantitative objectives drive continuous improvement.

Introduction

Learning Through Next-Word Prediction

Task Transformation Examples

Transforming Tasks into Next-Word Prediction

Core Idea: Reframe any task as continuing a text sequence.

- Provide context and prompt the model to generate the next sequence of words.
- Leverage language structure to complete various tasks.

Example 1: Detective Fiction

Task: Find the Real Murderer

Provide a narrative with clues and ask the model to continue the story:

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Task: Find the Real Murderer

Provide a narrative with clues and ask the model to continue the story:

> *"Detective Conan examined the clues—the broken vase, the misplaced candlestick, and the whispered alibi. Suddenly, it all clicked when he realized that..."*

The model's prediction could reveal the murderer by connecting subtle hints.

Example 2: Summarization

Task: Summarize a Document

Provide the text and prompt with:

> *"In summary, the main points of the article are..."*

The model predicts a concise summary.

Example 3: Translation

Task: Translate a Sentence

Given a source sentence and a prompt:

> *"Translate the following sentence into Spanish: 'The weather is nice today because the sun is shining brightly.' Translation:"*

The model outputs the translated sentence.

Example 4: Question Answering

Task: Answering Questions

Provide context and ask a question:

> *"Based on the passage, what is the main reason for the economic downturn? The answer is:"*

The model continues by predicting the answer.

Example 5: Code Generation

Task: Generate Code

Given a natural language description:

> *"Write a Python function that checks if a number is prime. def is_prime(n):"*

The model completes the code snippet.

Example 6: Creative Writing

Task: Story Continuation

Provide a partial story:

> *"In a world where magic and technology intertwine, the ancient prophecy began with the words 'When the moon eclipses the sun, the hidden power awakens,' and then..."*

The model predicts the story's continuation.

Why Does This Work?

- The next-word prediction framework taps into the model's deep understanding of language.
- It utilizes learned patterns, context, and knowledge to generate relevant continuations.
- This strategy makes LLMs flexible, capable of handling diverse tasks from creative writing to technical problem solving.